Laxmi Charitable Trust's

Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce

Department of Information Technology (B.Sc.I.T Semester IV)

MERN STACK

Practical – 2

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Class:TYIT	Batch:1
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A:Create a function and call them with Parameters.

CODE:

```
function calculateSalary(name, basicSalary){
    let bonus  = 5000;
    let totalSalary = basicSalary + bonus;

    console.log("employee name:", name);

    console.log("basic salary:", basicSalary);
    console.log("bonus:", bonus);
    console.log("total salary:", totalSalary);



}

calculateSalary("pratik", 30000);
console.log("pratik");
```

OUTPUT:

```
PS C:\Users\pratik mishra\OneDrive\Desktop\MERN STACK> node prac2a.js
employee name: pratik
basic salary: 30000
bonus: 5000
total salary: 35000
```

code:

```
function genBill(product,quantity,price){

    let total = quantity * price;

    let bill = total + (total*0.09);

    console.log("product", product);

    console.log("Quantity", quantity);

    console.log("Price per Item", price);

    console.log("total bill", bill);

    console.log("pratik mishra");


}


 genBill("lux", 2, 50000);
```

output:

```
PS C:\Users\pratik mishra\OneDrive\Desktop\MERN STACK> node prac2a.js
product lux
Quantity 2
Price per Item 50000
total bill 109000
pratik mishra
```

B: Work with array and object in java script.3

code:

```
let fruits = ["apple,", "banana","cherry","grapes"];

let students = {

    name: "pratik",

    age: 20,

    city: "mumbai"

};


console.log("fruits:", fruits);


console.log("fruits:", fruits[2]);



console.log("name:", students.name);

console.log("age:", students.age);

console.log("city:", students.city);

console.log("pratik");
```

output:

```
PS C:\Users\pratik mishra\OneDrive\Desktop\MERN STACK> node prac2b.js
fruits: [ 'apple,', 'banana', 'cherry', 'grapes' ]
fruits: cherry
name: pratik
age: 20
city: mumbai
pratik mishra
PS C:\Users\pratik mishra\OneDrive\Desktop\MERN STACK>
```

C: Demonstarte use of array function.

code:

```
let add = (a ,b) =>{

    return a+b;

}

    let sub = (a ,b) =>{

    return a - b;

    }


    let mul = (a ,b) =>{

    return a*b;

};


console.log("sum=", add(10,20));

console.log("sub=", sub(10,20));

console.log("mul=", mul(10,20));

console.log("pratik mishra")
```

output:

```
PS C:\Users\pratik mishra\OneDrive\Desktop\MERN STACK> node prac2c.js
sum= 30
sub= -10
mul= 200
pratik mishra
```

D: Write a program using array module map, filter reduce.

code:

```
let numbers = [10,20,30,40,50,];

let doubled = numbers.map(num => num*2);

let greater25 = numbers.filter(num => num>25);

let total = numbers.reduce((sum, num) => sum+num,0);

console.log("original array", numbers);

console.log("map", doubled);

console.log("filters", greater25);

console.log("reduce", total);

console.log(“pratik mishra”);
```

output:

```
PS C:\Users\pratik mishra\OneDrive\Desktop\MERN STACK> node prac2d.js
original array [ 10, 20, 30, 40, 50 ]
map [ 20, 40, 60, 80, 100 ]
filters [ 30, 40, 50 ]
reduce 150
pratik mishra
```

E:

Create a program that manipulate an object and display its output:

code:

```
let students = {

    name: "pratik",

    rollNo : 22,
```

```
    marks: 85
};
console.log("original object");
console.log(students);

console.log("accessing properties:");
console.log("name:", students.name);
console.log("marks:",students.marks);


students.marks = 90;
console.log("After updating marks:");
console.log(students);

students.city = "mumbai";
console.log("after adding city:");
console.log(students);

delete students.rollNo;
console.log("after deleting roll no:");
console.log(students);
```

output:

```
original object
{ name: 'pratik', rollNo: 22, marks: 85 }
accessing properties:
name: pratik
marks: 85
After updating marks:
{ name: 'pratik', rollNo: 22, marks: 90 }
after adding city:
{ name: 'pratik', rollNo: 22, marks: 90, city: 'mumbai' }
after deleting roll no:
{ name: 'pratik', marks: 90, city: 'mumbai' }
```

F:implement a small java script program combine function array and object mern.

code:

```
let students = [
    {
        rollNo: 101,
        name: "SOM",
        marks: 90
    },
    {
         rollNo: 102,
        name: "Pratik",
        marks: 90
    },
    {
        rollNo: 103,
        name: "Ashwin",
        marks: 90
    }
];
```

```
function display(s)
{
    console.log("student Details");
    console.log("------------------------");
    for(let stud of s )
    {
        console.log(`Roll No : ${stud.rollNo}`);
        console.log(`Name :${stud.name}`);
        console.log(`marks :${stud.marks}`);
        console.log("------------------------");
    }
}

display(students);
```

output:

```
PS C:\Users\pratik mishra\OneDrive\Desktop\MERN STACK> node prac2f.js
student Details
------------------------
Roll No : 101
Name :SOM
marks :90
------------------------
Roll No : 102
Name :Pratik
marks :90
------------------------
Roll No : 103
```

```
------------------------
Roll No : 103
Name :Ashwin
marks :90
```